

0.1 Charecteristics M. thau und M. bovis

	<i>Methanobrevibacter thaueri</i> CW ^T	<i>Methanobrevibacter boviskoreani</i> AbM4
form	coccobacillus	rods
G + C content (mol%)	38	29
T-Optimum	37 °C	38 °C
pH-Optimum	7	6.5-7
nutritional requirements	acetat and/or yeastextract/trypticase	yeastextraxt
growth on EtOH	—	+
growth on formate	—	+
antibiotic sensitiv	ND	chloramphenicol, puromycin *
antibiotic tolerance	ND	bacitracin, vancomycin *

*in *M. boviskoreani* JH^T

0.2 AB with organism

antibiotic	MIC [µg/ml]	organism	paper
Puromycin	2	<i>Methanococcus voltae</i>	Possot et al 1988
	5	<i>Methanosarcina mazei</i>	Mondorf et al 2012
Acetylpuromycin	100	<i>Methanococcus voltae</i>	Possot et al 1988
Neomycin	20	<i>Methanosarcina mazei</i>	Mondorf et al 2012
	125	<i>Methanobrevibacter arboriphilus</i>	Pecher & Böck 1988
	100	<i>Methanothermobacter thermautotrophicus</i>	Fink et al. 2023
Clarithromycin	0.25 - 16	<i>Mycobacterium abscessus</i>	Liao et al 2024
Chloramphenicol	25	<i>Methanobrevibacter smithii</i>	Dridi et al. 2011
	4	<i>Methanosphaera stadtmanae</i>	Dridi et al. 2011
Thiamphenicol	ND		
Bacitracin	4	<i>Methanobrevibacter. smithii</i>	Dridi et al. 2011

0.3 AB with resistance gen

antibiotic	MIC [$\mu\text{g/ml}$]*	effects	resistance gene
Puromycin	2;5	AA-tRNA-analoga -> attacks protein-biosynthesis	<i>pac</i>
Acetylpuromycin	100		product of <i>pac</i>
Neomycin	20;100;125	binds to 16s-Subunit -> misreading of mRNA	<i>neoR</i>
Clarithromycin	0.25;16	binds to 50s Subunit -> prevents transfer of amino acid chain	<i>ermB</i>
Chloramphenicol	4;25	binds to peptidyltransferase of 50s-Subunit -> prevents protein biosynthesis	<i>catP</i>
Thiamphenicol	ND	analoga of Chloramphenicol	<i>catP</i>
Bacitracin	4	inhibits transport of cellwall building-blocks	<i>BcrABC</i> ; <i>BacRS</i> ; <i>BacA</i>

*tested in close relatives